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Database Foundations

3-2

Tracking Data Changes

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Objectives

- This lesson covers the following objective:
 - Keep track of data that changes over time



Model Data over Time: Accommodating Historical Data

- Most businesses need to track some historical data to help them find trends and patterns that can be the basis for business innovations or process improvements
- Every update of an attribute or transfer of a relationship means loss of information. Often information is no longer useful, but some systems must keep track of some or all of the historical values of an attribute. In addition, some systems are required to keep an audit trail of each transaction
- Validate any requirements for storing historical data with the user, because storing unnecessary historical data can be costly

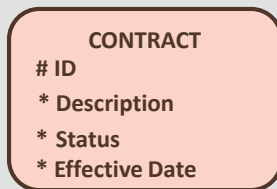
Model Data over Time: Accommodating Historical Data

- Add entities and relationships to the model to accommodate historical data
 - Is an audit trail required?
 - Can attribute values change over time?
 - Can relationships change over time?
 - Do you need to query older data?

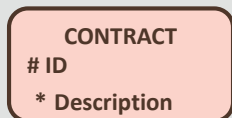


Model Data over Time: Tracking Attribute Values

- Create additional entities to track the attributes' values over time

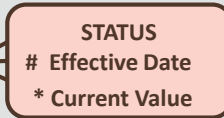


Use one entity if you track only the current status of a contract



contain

state of



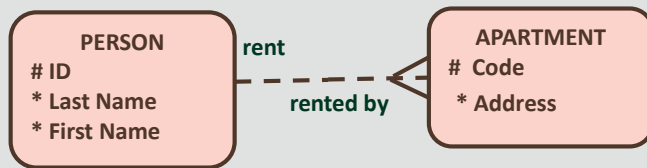
Add an entity to store the status of a contract over time

As shown in the slide example, the **CONTRACT** entity was initially created with four attributes. Because the attributes are contained in one entity, the contract's status and effective date will change to the current status and effective date each time the contract changes. If you are required to track *when* the status of a particular contract changes, create an additional entity and move the status and effective date to the new entity. In this way, the information about when the status of a contract changed can be tracked.

If the status changes more than once on the same day, you should include a current status value attribute in the unique identifier.

Model Data over Time: Accommodating a Changing Relationship

- Add an entity to accommodate a relationship that may change over time

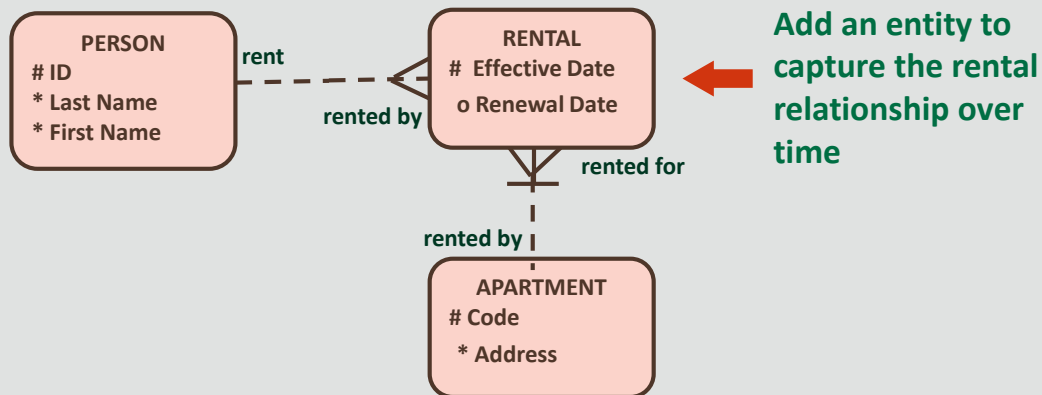


This does not allow tracking of the date that an apartment is rented.

You may need to add an entity to keep track of how a relationship changes over time. In the slide example, a person can rent an apartment, but an apartment can be vacant. If you want to track which apartments are rented at a particular point in time, add an entity to store rental information about an apartment.

Model Data over Time: Accommodating a Changing Relationship

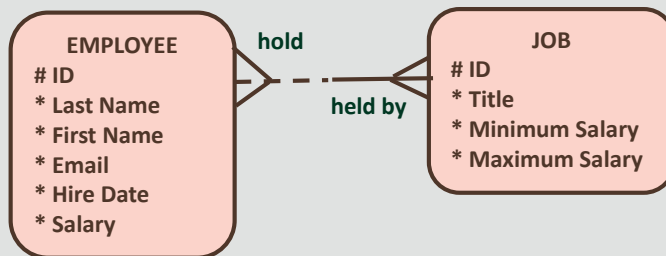
- Add an entity to accommodate a relationship that may change over time



In the RENTAL entity, the apartment code and rental effective date are the unique identifier. You would not include the person ID because you are storing the current and historical rentals. You would not have two rentals for the same apartment with the same effective date.

Model Data over Time: Creating Intersection Entities

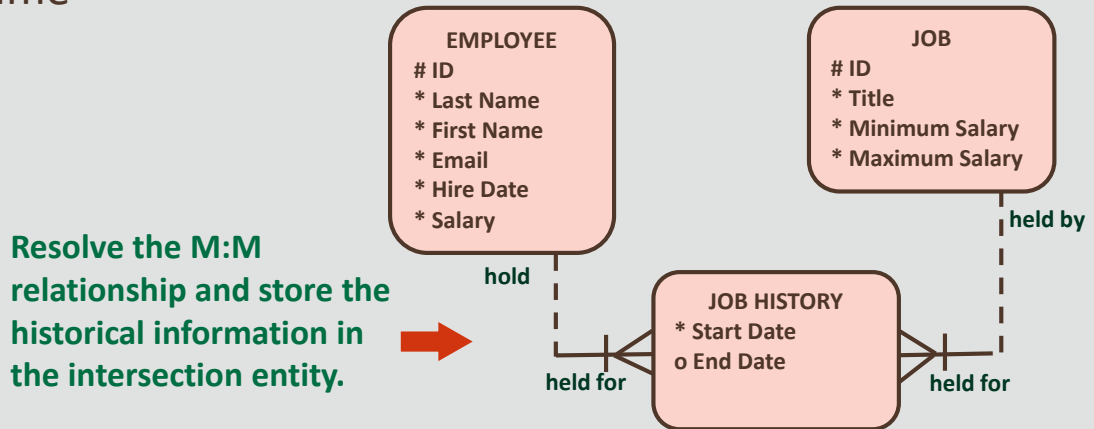
- Resolving M:M relationships with an intersection entity is frequently used to track information about a relationship that changes over time



← This does not store the history of an employee's jobs

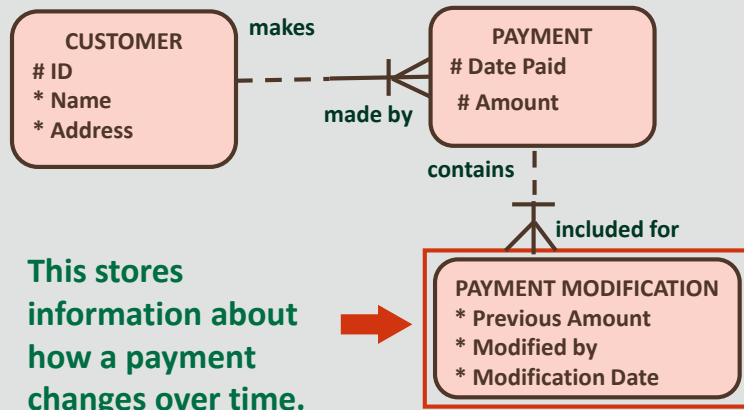
Model Data over Time: Creating Intersection Entities

- An intersection entity is frequently used to track information about a relationship that changes over time



Model Data over Time: Logging Modified Information

- Additional entities and attributes store modified information

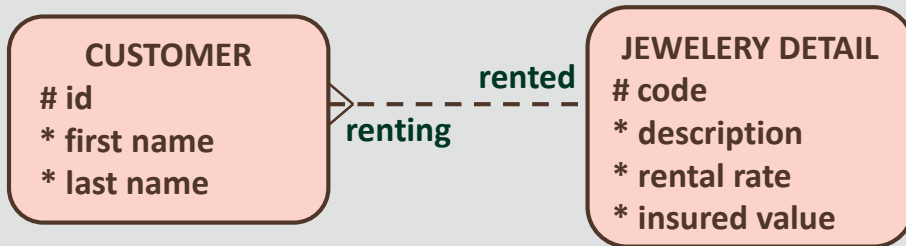


When a system allows a user to modify or remove particular information, you may decide to keep the old values. This is called logging or journaling and is common in financial type models.

In the slide example, a customer may make multiple payments and you can make changes to a payment. If you want to track those changes, you can store the history in the PAYMENT MODIFICATION entity.

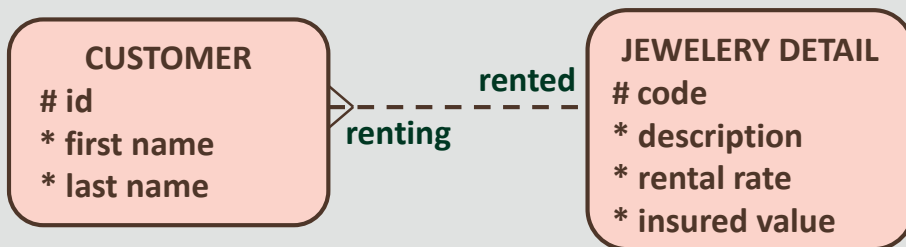
Model Rental over Time: Scenario

- A jewelry store rents necklaces, bracelets, and the like for special occasions, such as award ceremonies and movie premieres
- The store wants to track the rental history



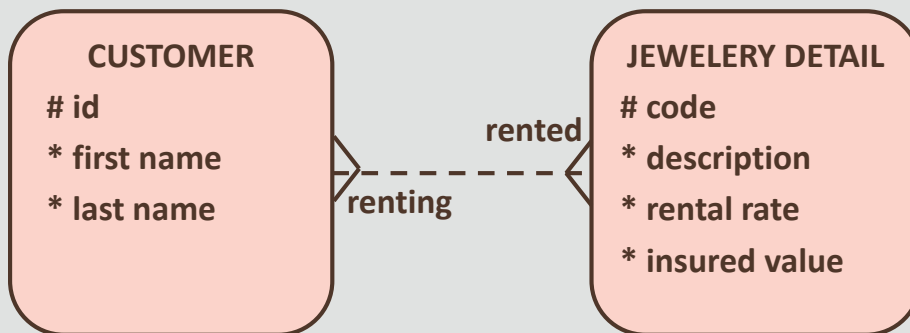
Model Rental over Time: Scenario

- The ERD model tracks only the current renter
- How would you revise the relationship to track the history?



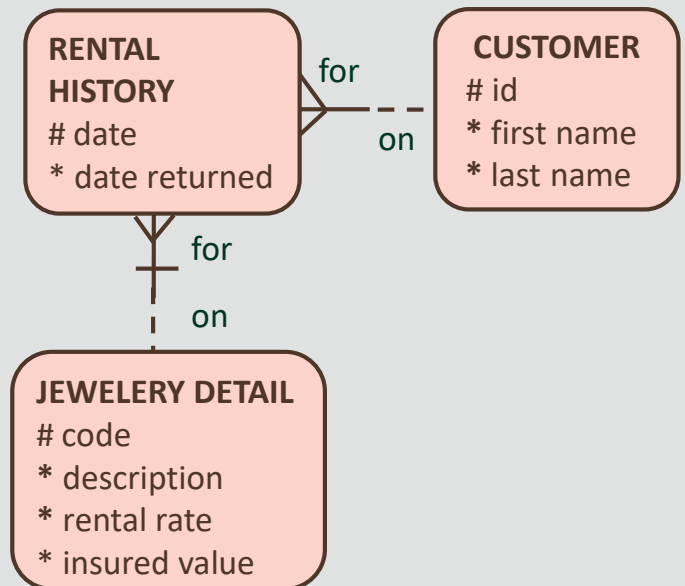
Model Rental over Time: Scenario

- Change the relationship between JEWELRY DETAIL and CUSTOMER to M:M.



Resolve M:M

- Resolve M:M with the RENTAL HISTORY intersection entity
- What is the UID of RENTAL HISTORY?
- What piece of information is not being tracked?



The UID of RENTAL HISTORY is the date rented and the UID of JEWELRY DETAIL (shown by the barred relationship). Customer id is not being tracked in RENTAL HISTORY.

Project Exercise

- DFo_3_2_Project
 - Oracle Baseball League Store Database
 - Keeping Track of Data that Changes over Time



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DFo 3-2
Tracking Data Changes

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Summary

- In this lesson, you should have learned how to:
 - Keep track of data that changes over time



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